

- 3 (a) (i) *either* sum / total momentum (of system of bodies) is constant
or total momentum before = total momentum afterM1
for an isolated system / no (external) force acts on system A1 [2]
- (ii) zero momentum before / after decayM1
so α -particle and nucleus D must have momenta in opposite directions A1 [2]
- (b) (i) kinetic energy = $\frac{1}{2}mv^2$ C1
 $1.0 \times 10^{-12} = \frac{1}{2} \times 4 \times \underline{1.66} \times 10^{-27} \times v^2$ M1
 $v = 1.7 \times 10^7 \text{ m s}^{-1}$ A0 [2]
- (ii) $1.7 \times 10^7 \times 4u = 216u \times V$ C1
 $V = 3.1 \times 10^5 \text{ m s}^{-1}$ A1 [2]
(accept $3.2 \times 10^5 \text{ m s}^{-1}$, do not accept 220 rather than 216)

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- (c) $(1.7 \times 10^7)^2 = 2 \times \text{deceleration} \times 4.5 \times 10^{-2}$
 deceleration / $a = 3.2 \times 10^{15} \text{ m s}^{-2}$
 (accept calculation based on calculating $F = 2.22 \times 10^{-11} \text{ N}$
 and then use of $F = ma$)

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